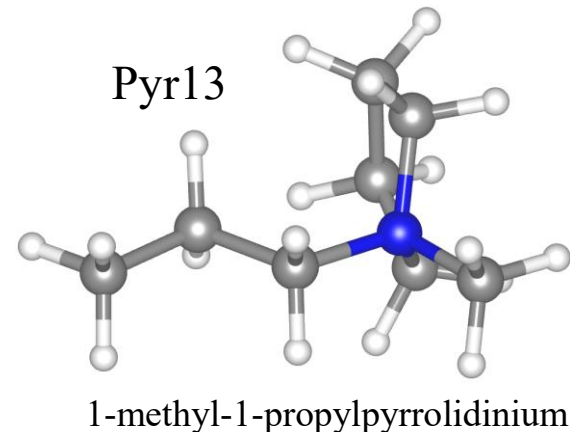
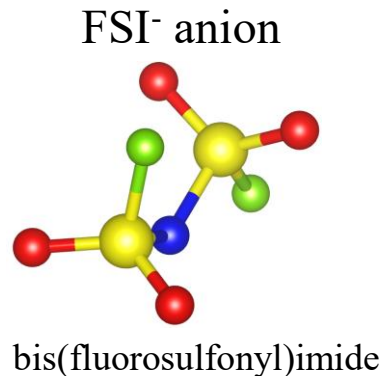
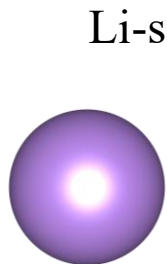


Lecture 1 -Introduction to Ionic Liquid Electrolytes

Overview and structure of Concentrated Ionic Liquid Electrolytes(CILEs)

- Ionic liquids are salts with a melting point below 100 °C.
- They are composed of anions and cations; no water is present.
- Their regular and ordered structure gives them very different properties from conventional liquids.
- Ionic liquid based on FSI-Pyr13 has been well studied as a novel electrolyte.



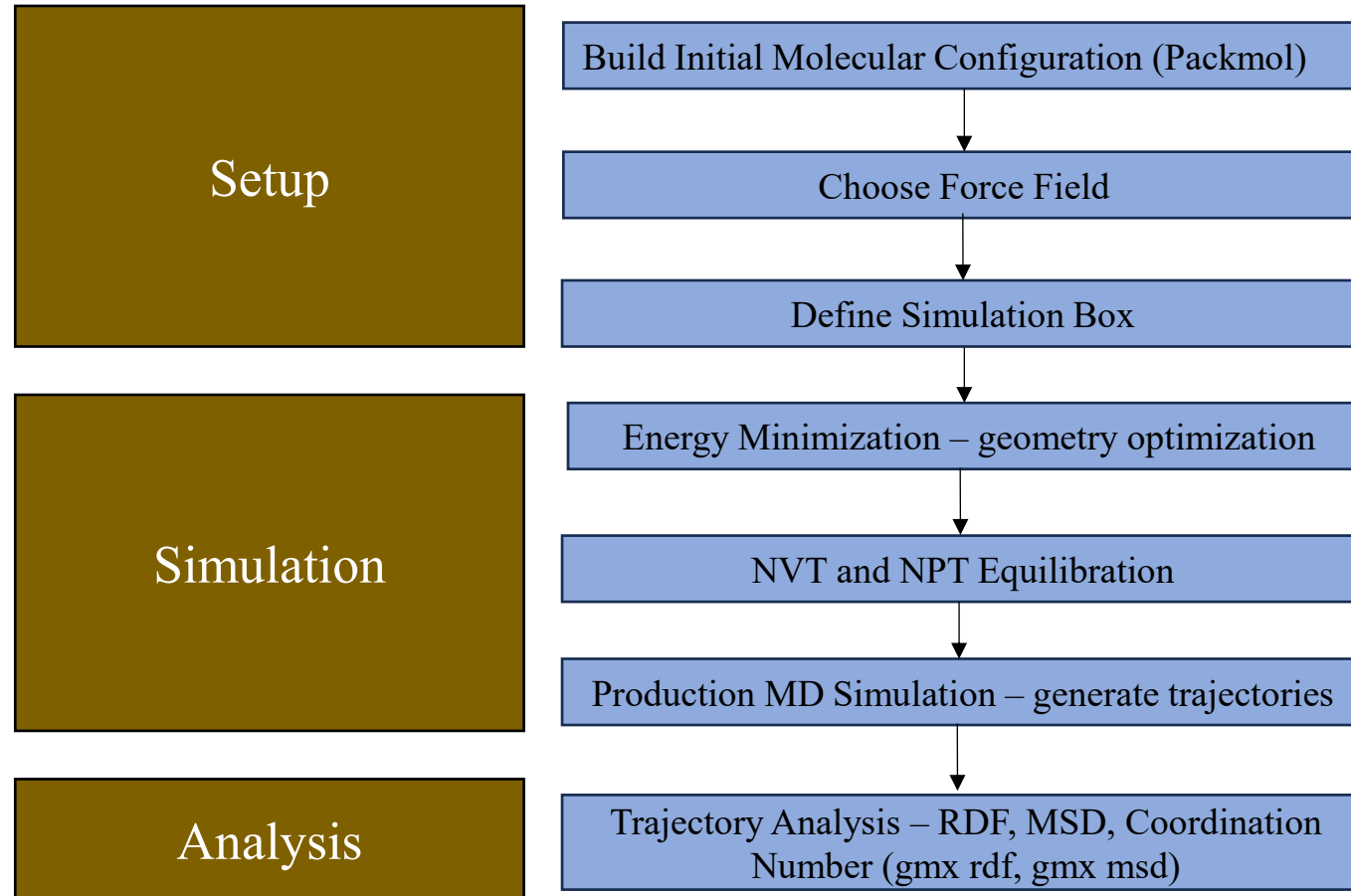
We will use three concentrations CILEs comprising Lithium bis(fluorosulfonyl)imide (LiFSI) as the primary salt, 1-methyl-1-propylpyrrolidinium bis(fluorosulfonyl)imide (Pyr₁₃FSI) as a pyrrolidinium-based IL of in this study.

- 1) 1 LiFSI, 2 Pyr₁₃FSI denoted as L1P2
- 2) 2 LiFSI, 2 Pyr₁₃FSI denoted as L2P2
- 3) 3LiFSI, 2 Pyr₁₃FSI denoted as L3P2

Molecular Dynamics Simulations

- Perform **molecular dynamics (MD) simulations** using the GROMACS package.
- Characterize the **local solvation environment** by calculating the **radial distribution function (RDF)** and **coordination number (CN)**.
- Evaluate **ionic transport properties** through **mean square displacement (MSD)** analysis and the corresponding **diffusion coefficients**.
- Finally, assess **structural stability and interaction strength** using **density functional theory (DFT)** calculations by determining the **binding energies** of key species.

Molecular Dynamics Simulation Framework



Tutorial Packmol

Thank you